Supervised Fine-tuning for Gen Z Slang Summaries

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Abstract

Traditional summarization systems produce factually accurate but stylistically neutral outputs, limiting their appeal to audiences that employ contemporary online slang. This project presents a supervised fine-tuning approach for abstractive summarization with demographic-specific style transfer targeting Gen Z audiences. We construct an enhanced Gen Z slang dictionary dataset by merging three public datasets (1,888 unique terms) and use an augmented TL;DR corpus containing 116,578 synthetically generated slang-formal sentence pairs via semantically-guided LLM rewriting. Using this corpus, we fine-tune T5-small and BART-base models on Reddit TL;DR summaries to generate stylistically authentic Gen Z-style summaries. Evaluation of the two models shows that BART is superior in all measured categories. We demonstrate that pretrained summarization models can be effectively adapted for generation-specific style transfer through carefully constructed synthetic training data and supervised fine-tuning. Our dual evaluation framework combining automatic metrics with LLM-based judgment provides robust assessment of both semantic preservation and stylistic authenticity.

1 Introduction

Younger generations' consumption of informal linguistic styles is shaped by social media, memes, and online communities [1]. Gen Z, in particular, uses mainly acronyms, imitative expressions and flippant slang, which rapidly evolve and differ from the neutral language typically produced by automated summarization systems [2]. While modern abstractive summarization models have advanced significantly, their outputs remain optimised for clarity and factual correctness rather than demographic-specific communication styles. As a result, summaries often lack the engagement, relatability, and authenticity necessary to resonate with Gen Z users. Therefore, creating tools that bridge and adapt text to summaries in different styles can significantly enhance the effectiveness, appeal, and satisfaction of communication [3, 4, 5].

This disconnect in style between the text and the summary is amplified by the absence of comprehensive, high-quality Gen Z slang datasets. Existing resources exhibit challenges such as incomplete coverage, duplication of slang words, limited contextual information, or a lack of direct formal-to-slang mappings. These limitations make it difficult to train models that can reliably perform style transfer into contemporary youth language. Furthermore, prior work in text style transfer has focused primarily on broad stylistic dimensions such as formality, sentiment, or politeness, leaving modern slang and generation-specific communication largely unexplored [6, 7].

To address these limitations, this project combines summarization and style transfer into a unified pipeline capable of producing concise, yet stylistically engaging summaries. Our contributions include the construction of a robust merged Gen Z dictionary derived from three datasets and augmented with synthetic examples, as well as the expansion of the TL;DR summarization corpus with a new Gen Z summary column, yielding 116,578 AI-generated samples that we have used for fine-tuning pretrained summarisers. Through supervised fine-tuning (SFT) of a T5-small and a BART-base model, we investigate whether these architectures can learn informal style transfer patterns from limited training data and which one performs better.

2 Related Works

We utilize multiple complementary datasets to address the lack of robust Gen Z slang resources.

Programmer-RD-AI/genz-slang-pairs-1k [8]: provides paired normal and Gen Z slang sentences. Although it lacks explicit slang word annotations, these can be extracted algorithmically to correct and complete the slang dictionary for more accurate explanations.

MLBtrio/genz-slang-dataset [9]: contains Gen Z slang terms with descriptions, context, and usage examples intended for fine-tuning slang models. However, it lacks direct translations to standard English. We augment this dataset with a new column translating slang terms into neutral English, aligning with the genz-slang-pairs-1k format.

Tawfiayeasmin/gen-z-words-and-phrases-dataset [10]: provides a Gen Z slang dictionary with popularity scores. This dataset contains mostly unique entries with minimal overlap with other datasets, making it valuable for extending our slang dictionary. However, the popularity metric cannot be effectively utilised due to inconsistent support across datasets.

TL;DR [11]: contains over 100,000 Reddit post-summary pairs for extreme summarization. We extend this corpus with a new column of synthetically generated Gen Z-style summaries to create paired training data for supervised fine-tuning.

3 Methods

3.1 Generating Synthetic Data

Our analysis of existing datasets revealed that no single resource is sufficiently robust for our task. Therefore, we combine and enhance three complementary Gen Z slang datasets using LLM-based augmentation to generate synthetic training data. For details, see the jupyter notebook *data_generation.ipynb* [12]. The synthetic TL;DR data generation process consists of following.

1. Creating embeddings for slang terms. We generate sentence embeddings for the neutral English translations of slang terms from our combined Gen Z dictionary. These embeddings capture the semantic meaning of each slang concept, enabling similarity-based retrieval. We chose to embed neutral text rather than slang descriptions because neutral translations provide clearer, more consistent semantic representations that align better with the formal language in TL;DR completions. Descriptions often contain usage examples and contextual information that introduce noise into the embedding space.

2. Embedding TL;DR completions. Each completion (summary) from the TL;DR dataset is embedded at the post level. Post-level embeddings capture the overall semantic content of the entire summary, providing a holistic representation that facilitates matching with relevant slang concepts. Sentence-level embeddings would fragment the semantic context, while document-level embeddings (of full Reddit posts) would introduce excessive noise from content that does not appear in the summary. By embedding at the completion level, we ensure that retrieved slang terms are semantically relevant to the summary content.

3. Top-k slang retrieval. For each TL;DR completion embedding, we compute cosine similarity with all slang term embeddings and select the top-k most similar terms. We set $k = 8$ based on the rationale that incorporating 8 slang terms into a 1-to-3 sentence summary provides sufficient stylistic richness without overwhelming the text. A smaller k risks losing important stylistic elements, while a larger k could result in forced or unnatural slang usage.

4. LLM-based rewriting. The original TL;DR completion, along with the top 8 retrieved slang terms and their descriptions, is provided to an LLM with instructions to rewrite the summary in Gen Z style. The prompt explicitly directs the model to preserve semantic meaning while naturally incorporating slang terms without overuse or forced insertions. This approach ensures that the synthetic Gen Z summaries maintain factual accuracy and coherence while adopting appropriate stylistic attributes.

We have generated 116,578 synthetic completion pairs. Supervised fine-tuning typically does not require millions of examples, prior work suggests that 100 to several thousand high-quality examples suffice for effective SFT on specific tasks. We split the dataset into 20,000 samples for training, 3000 for validation, and 3000 for testing. This subset of the data is of reasonable size to achieve fine-tuning results while not being computationally too expensive for the time-scope of the project.

3.2 Model Architecture and Training

Our objective is to perform supervised fine-tuning of a pre-trained summarization model to generate Gen Z-style summaries. Our training input consists of the original TL;DR posts, target labels are Gen Z completions and for the input prompt uses a prefix string *summarize with Gen Z slang:* . During fine-tuning, the model learns to map from the original post directly to the Gen Z-style summary.

We explore two sequence-to-sequence architectures.

T5-small [13]: a 60.5M parameter encoder-decoder model trained on diverse text-to-text tasks. T5-small is computationally efficient and suitable for resource-constrained environments.

BART-base [14]: a 139M parameter denoising autoencoder that combines BERT-style bidirectional encoding with autoregressive decoding. BART's asymmetric architecture is well-suited for abstractive summarization.

We employ identical hyperparameters across both models for fair comparison: optimization (AdamW with 8-bit precision to reduce memory consumption); learning rate (2×10^{-5} with cosine annealing and 6% warmup); regularization (weight decay of 0.001, gradient clipping at 0.3); training (10 epochs with a batch size of 4, 1 per device with 4 gradient accumulation steps); generation (beam search with 4 beams during validation; a repetition penalty of 1.5, and a no-repeat-ngram-size of 3); memory optimization (gradient checkpointing, mixed precision (fp16); torch cache clearing every step).

3.3 Evaluation Methodology

To evaluate the performance of the model, we have implemented two evaluation methods: automatic metrics and an AI judge using a GPT-based LLM.

First, we are creating our custom metric, which is a combination of a BERTScore F1 and a slang density, which is a ratio of slang words used in a generated summary. We want to ensure that the distributions of these match our baseline, the synthetic data in the TL;DR corpus. Table 1 shows the results.

The second method uses an LLM as a judge, where we prompt it to evaluate three categories of the generated summaries: meaning preservation (does it keep core facts, details, and original intent?), slang quality (is Gen Z slang natural, fluent, and appropriately used?), and reddit-style naturalness (does it sound like a real redditor's TL;DR?). All of them receive a score between 1-5, where 5 is the best. We also get a combined overall score, which is a sum of the three judged categories.

3.3.1 Automatic Metrics

Style Fidelity Score: the overall style fidelity metric combines two components:

$$\text{style\_fidelity} = 0.90 \times \text{bertscore\_f1} + 0.10 \times \text{slang\_fidelity}\,. \tag{1}$$

Semantic Quality (90% weight): measured using BERTScore F1, which evaluates the contextual similarity between synthetically generated and original reference summaries. This ensures that the generated text preserves the meaning of the input's TL;DR prompt.

Slang Fidelity (10% weight): evaluates whether the model maintains slang density consistent with authentic reference data. The baseline statistics computed over the training set (mean = 0.0814, std = 0.0492) define expected slang usage ($\sim$8% of words are slangs), with a $\pm 2\sigma$ tolerance band to account for natural linguistic variation without penalising reasonable divergence.

The slang fidelity metric specifically measures deviation from the baseline of the TL;DR synthetic completions, ensuring the model's output will have a reasonable amount of slang:

$$\text{slang\_fidelity}_i = \max\left(0.0, 1.0 - \frac{|\text{pred\_density}_i - \text{baseline\_mean}|}{2 \times \text{baseline\_std}}\right)\,. \tag{2}$$

4 Results

4.1 Training Dynamics

Figure 1 depicts training curves for both models, showing convergent behaviour over 10 epochs. T5-small achieves a final validation style fidelity score of 0.5528, with training loss decreasing from 3.484 to 2.952. BART-base achieves a marginally higher validation style fidelity of 0.5637, with training loss decreasing from 3.485 to 2.726.

Both models demonstrate stable training with gradually decreasing validation loss, indicating appropriate hyperparameter selection and the absence of overfitting. The modest improvement in style fidelity scores across epochs suggests that the models have adequately learnt the style transfer patterns from the training data.

Table 1: Comparison of BART and T5 Models on Style Fidelity, BERTScore F1, and Slang Fidelity.

Metric	BART	T5	Baseline
Style Fidelity	**0.5637**	0.5543	NA
BERTScore F1	0.5587	0.5441	**0.6333**
Slang Fidelity	0.6087	**0.6459**	NA

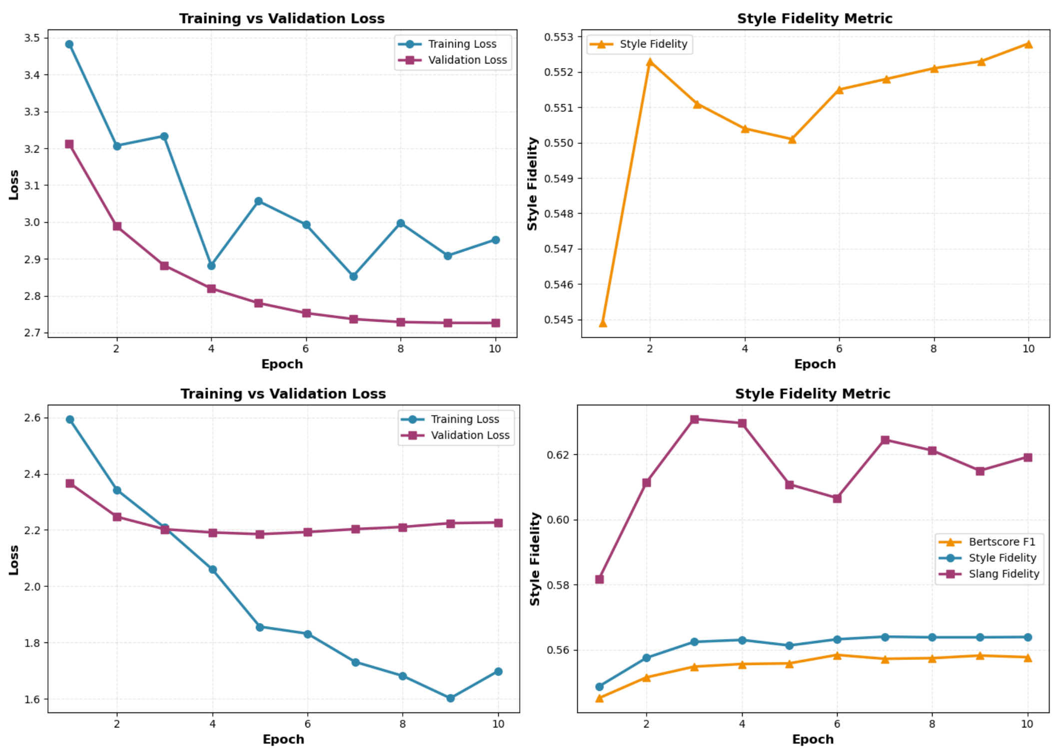

Figure 1: Training and validation over 10 epochs. T5 (top) and BART (bottom).

4.2 LLM-based Comparative Evaluation

To validate automatic metrics and obtain human-interpretable judgments, we conduct a comparative evaluation using an external LLM as a judge (openai.gpt-oss-120b-1:0 model on Amazon Bedrock). We evaluate 3,000 test instances using three 1-5 scoring dimensions. Table 2 shows the results.

BART-base achieves the highest overall score (7.664 out of 15), exceeding the baseline synthetic reference (7.402) and substantially outperforming T5-small (6.565). Notably, BART-base achieves the highest scores in slang quality (2.709 vs 2.499 baseline) and Reddit naturalness (2.752 vs

2.578 baseline), indicating that the model produces stylistically authentic Gen Z summaries. The improvement in slang quality suggests that BART-base has learnt to incorporate slang naturally and contextually. The meaning preservation is very close to the baseline (2.203 vs 2.325).

T5-small's lower performance across all three dimensions is particularly concerning for meaning preservation (1.723 vs 2.325 baseline), indicating that T5-small struggles to maintain semantic fidelity while performing style transfer. This suggests that T5-small's smaller parameter capacity (60.5M vs 139M for BART-base) limits its ability to simultaneously learn style transfer and maintain content fidelity.

Table 2: Performance Comparison of Baseline, T5, and BART Models across Meaning, Slang, and Reddit Categories.

Model/Source	Meaning	Slang	Reddit	Overall
Baseline	**2.325**	2.499	2.578	7.402
T5	1.723	2.427	2.415	6.565
BART	2.203	**2.709**	**2.752**	**7.664**

4.3 Qualitative Analysis

To gain deeper insights into model behaviour, we examine qualitative examples comparing baseline synthetic references, T5-small outputs, and BART-base outputs. These examples align with the quantitative findings where BART-base demonstrates a superior ability to perform simultaneous style transfer and semantic preservation compared to T5-small.

Table 3 shows T5-small's failure to perform style transfer, it reverts to neutral language despite explicit prompt instructions. BART-base successfully incorporates slang ("sus," "bread," "chill") while preserving meaning. Similarly, Table 4 shows T5-small's forced and awkward slang integration ("not wrong fr fr"), whereas BART-base produces more natural, flowing Gen Z language while preserving the core dilemma and its context.

Table 3: Workplace Conflict Resolution. Comparison of baseline synthetic reference, T5-small, and BART-base outputs on a Reddit post about working overtime without compensation. T5-small reverts to neutral language entirely, while BART-base naturally incorporates Gen Z slang ("sus," "bread," "chill") while preserving meaning and emotional tone.

Source	Output Text
TL;DR Post	"My boss asked me to work on Saturdays for the next three months without overtime pay. Should I be concerned?"
Baseline	"So your boss wants you to grind Saturday shifts for three months with no extra cash. That's lowkey sus, not okay at all."
T5-small	"Your boss wants you to work Saturdays for three months without overtime pay." (drops slang entirely)
BART-base	"Your boss is making you pull Saturday shifts for three months with zero overtime bread. That's mad sus and definitely not chill."

5 Discussion

The consistent superiority of BART-base across automatic and LLM-based metrics can be attributed to several architectural and capacity factors. Firstly, BART's asymmetric encoder-decoder architecture provides inherent inductive biases favourable for abstractive summarization. The bidirectional encoder captures rich contextual representations, while the autoregressive decoder enables flexible, controllable generation. T5's symmetric encoder-decoder architecture, while more general-purpose, may not provide the specialised inductive bias needed for simultaneous content preservation and style transfer.

Additionally, the parameter count difference (139M vs 60.5M) is substantial. BART-base's 2.3× larger model capacity provides greater representational flexibility to learn both the semantic content of

Table 4: Relationship Commitment Dilemma. Comparison of outputs on a Reddit relationship advice post. T5-small exhibits dropped content and missing slang patterns, while BART-base produces coherent Gen Z language that maintains narrative flow, emotional tone, and semantic content.

Source	**Output Text**
TL;DR Post	"I've been dating my girlfriend for 5 years. Her family thinks I should propose by now, but I'm not ready. Am I wrong?"
Baseline	"Been with my girl for five years and her fam is lowkey pressuring me to pop the question. I'm not feeling it yet, no cap."
T5-small	"I've been dating my girlfriend for 5 years but her family thinks I should propose." *(drops slang entirely)*
BART-base	"Been with the same girl for five years now but her fam keeps pestering me to propose. I'm just not there yet, fr fr no cap."

summaries and the stylistic patterns of Gen Z language. The capacity constraint in T5-small appears to force a trade-off: when instructed to incorporate Gen Z slang, T5-small partially sacrifices semantic fidelity. This trade-off is evident in Table 3, where T5-small abandoned style transfer entirely to preserve meaning.

Lastly, BART's pretraining regime emphasises text generation quality through its denoising autoencoder objective, which may better prepare the model for controlled generation tasks. T5's text-to-text framework, while versatile, optimises for diverse tasks without specialising in abstractive generation quality.

5.1 Limitations and Baseline Quality

A critical observation is that the synthetic baseline itself achieves only moderate scores (7.402 overall), suggesting that the synthetic reference data used for training may limit the model quality. The LLM-generated synthetic summaries contain errors such as missing key information details, occasional forced slang usage, and inconsistent style. This data quality ceiling constrains how well fine-tuned models can perform. While BART-base exceeds the baseline (7.664 vs 7.402), the improvement of 3.5% indicates that the models have learnt from imperfect training signals.

Future work should employ an iterative data curation process: generate synthetic summaries, score them using the LLM judge, filter only high-quality examples (e.g., overall score > 11/15), and use this filtered subset for fine-tuning. This approach would produce a higher-quality "gold standard" baseline and likely enable better model performance.

6 Conclusion

We have presented a supervised fine-tuning approach for generating Gen Z-style summaries by combining three existing Gen Z slang datasets and augmenting them with synthetically generated examples. Our enhanced Gen Z dictionary merges 1,888 unique slang terms across three sources, providing comprehensive coverage of contemporary youth language. Through a semantically-guided embedding approach, we generate 116,578 synthetic training examples that pair Reddit TL;DR summaries with Gen Z-styled variants, enabling effective supervised fine-tuning of pretrained sequence-to-sequence models.

A comparative evaluation of T5-small and BART-base reveals that BART-base achieves superior performance across multiple evaluation dimensions. Automatic metrics show that BART-base achieves 2.0% higher BERTScore F1 (0.5587 vs 0.5441) and better overall style fidelity (0.5637 vs 0.5543). LLM-based evaluation confirms this advantage, with BART-base achieving an overall score of 7.664 compared to T5-small's 6.565 and the baseline synthetic reference's 7.402. Notably, BART-base excels at slang quality and Reddit naturalness, producing authentic Gen Z language while preserving semantic content.

The work demonstrates that existing summarization models can be effectively adapted for demographic-specific style transfer through carefully constructed training data and supervised fine-tuning, and that the dual evaluation approach provides a robust assessment of both semantic and stylistic dimensions of the model's output.

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